

Basic sound setup

About **\$430** with a borrowed calibrator; **\$670** with our own.

What this buys

- Use our existing computer and brain-recording system.
- Play selected tones below **20 kilohertz**. A higher frequency means a higher-pitched sound.
- A microphone near the ear checks the sound that actually arrives. A sound-level calibrator produces a known test sound to check the microphone.

Computer → **sound interface** → **powered speaker**

Microphone near ear → **sound interface** → **sound recording**

Parts to buy

Blue names are buying links. Exact product codes are kept for ordering; explanations use full words. Prices are in United States dollars, checked October 9, 2026.

Job	Part / seller	Price
Play and record	Behringer UMC204HD – eBay / Sweetwater, new	97.98
Make the sound	One JBL 305P MkII powered speaker – Sweetwater, new/sale	99.00
Measure sound	Dayton EMM-6 microphone – eBay / Parts Express, new	66.93
Microphone cable	Hosa HMIC-010 microphone cable – Sweetwater	17.99
Audio cables	Two Hosa HSS-005 audio cables , \$14.95 each	29.90
Microphone stand	Gator RI-MICTP-FBM stand – Sweetwater	34.99
Parts	Borrow the sound-level calibrator	346.79
Optional purchase	REED R8090 sound-level calibrator – DigiKey / REED, new	219.00

- **Borrow the calibrator:** \$346.79 parts + \$80 estimated shipping/tax = **\$426.79**.
- **Buy the calibrator:** \$565.79 parts + \$100 estimated shipping/tax = **\$665.79**.
- **Still extra:** any cable or adapter needed to connect timing signals to our brain recorder.

What could be misaligned?

- **Start delay:** storing sound in a computer buffer adds delay. As an example, one buffer of 256 samples at 48,000 samples per second lasts **5.3 milliseconds**; two last 10.7 milliseconds. These are calculations, not measurements of this interface.
- **Sound travel:** the speaker and converters add delay. Air adds about **3 milliseconds per metre** from speaker to ear.
- **Loudness accuracy:** unknown until each tone is checked at the ear.
- **Changing delay between trials:** unknown until we test it. Preload the sound sequence and use wired connections.
- **Drift:** the sound and brain recorders may run at slightly different speeds. Our mismatch is unknown. For scale only, a hypothetical 0.01% speed mismatch would cause **0.36 seconds of error in one hour** without correction.

How we check the timing

- Send a timing reference beside each tone and record that reference with the brain signals.
- Use the microphone to measure the gap between the reference and sound arriving at the ear.
- Recording the microphone only on the sound interface leaves two timelines to align. A sound output is not automatically a compatible digital trigger; check the recorder input.

Sources: [computer buffering](#); [delay estimates](#); [sound travel](#). Sound level must also be checked for each tone at the ear.

Accurate version 1: SONIC paper

\$4,840.36 priced so far; speaker/amplifier quotes and measurement equipment are extra.

What this buys

- Reproduce the paper’s sound chain: **sound device** → **speaker amplifier** → **speaker**.
- Add a satellite-based timing reference to help coordinate equipment clocks.
- Reuse our brain recorder. The paper’s Connexus recorder is outside this budget; its reported processing delay cannot be inherited by buying these sound parts.

Parts to buy

One of each. Prices are in United States dollars, checked October 9, 2026. National Instruments makes the sound device; Tucker-Davis Technologies makes the speaker/amplifier. “Quote” means ask the supplier for a price.

Part	Seller	Price
USB-6361 sound device, 782255-01	New / Newark	3,744.42
TimeMachines TM2500C time reference	New / backorder	799.99
MF1-M speaker kit	Supplier quote	Quote
SA1 speaker amplifier	Supplier quote	Quote
ZB1PS powered enclosure*	Supplier quote	Quote
781513-01 power supply*	New / Newark	123.42
780534-01 computer cable*	New / DigiKey	69.75
763830-01 United States power cord*	New / Newark	51.00
781233-02 noise filter*	New / DigiKey	38.25
Hosa BNC-58-103 output lead*	New / Pro Audio Solutions	13.53
Subtotal of priced parts		4,840.36

What is still extra?

- **Sound measurement:** our page-4 kit with a new microphone amplifier adds \$2,517.22. Priced parts then total **\$7,357.58**, before supplier quotes.
- **Connections:** microphone-to-sound-device, network and brain-recorder wiring still need selection. The timing reference is currently backordered.
- **Accessories:** starred parts are our reproduction additions, not exact accessories named by the paper. Confirm included cables, stands and power supplies to avoid buying duplicates.

How precise is it?

- **Pitch range:** the paper tested **1.4–32 kilohertz**. It used about 80 decibels at the ear, but gives no measured sound-level uncertainty.
- **Reported neural-processing delay: 56 milliseconds.** This covers neural measurement, filtering and collecting data for decoding. It excludes sensory processing and computer calculation time. A separate one-session analysis reports **11 milliseconds**.
- **Sound-to-brain recording alignment:** remaining timing error and long-session drift are not quantified. The time reference specifies below one microsecond plus network variation, not sound-arrival accuracy.
- **Drift:** the independent sound-device clock specification corresponds to **180 milliseconds per hour versus ideal time**. This is not the paper’s drift with satellite timing connected. Drift against our recorder remains unknown.

Sources: SONIC version 1, pages 5, 7–9 and 11; sound device specifications; time reference; speaker. The earlier \$3,498.95 eBay lead needs current availability/model confirmation. Reported 56 milliseconds: neural measurement below one, filtering five, binning five and context 45. Clock figure is typical at 25 degrees Celsius. The paper used awake sheep; Rice’s sedated-sheep adaptation remains unverified. No complete price or upper limit is established.

Accurate version 2: McGinley slides

\$7,551.75 priced so far; speaker quote and measurement equipment are extra.

What this buys

- Reproduce the supplied setup: **computer** → **equipment rack** → **speaker driver** → **speaker**.
- The rack holds a sound board and a separate signal input/output board.
- Reuse our brain recorder instead of buying the slides' Neuropixels system. A compatible desktop computer with a free expansion slot is required.

Parts to buy

One of each. Prices are in United States dollars. Most rack parts are used; several ship from China. Seller claims do not verify operation or calibration.

Part	Seller	Price
PXIe-1082 rack enclosure	Used / eBay	2,650.00
PXIe-6363 signal input/output board	Used / eBay	2,495.36
PXI-4461 sound board, 186900-11L	Used / eBay	699.99
PXIe-8360 rack controller	Used / eBay	493.20
PCIe-8361 computer connection card	Seller-new / eBay	197.00
779500-03 rack cable, 3 metres	Seller-new / eBay	89.85
SHC68-68-EPM cable, 192061C-02	Used / eBay	89.99
BNC-2090A connection panel	Used / no returns	613.77
ED1 driver + enclosure/power supply	Used / eBay	209.06
ES1 speaker + matching cable	Supplier quote	Quote
Hosa BNC-58-103 output lead*	New / Pro Audio Solutions	13.53
Subtotal of priced parts		7,551.75

What is still extra?

- **Sound measurement:** our page-4 kit with a new microphone amplifier adds \$2,517.22, making **\$10,068.97** priced so far. The slides do not identify a microphone or sound-level calibrator.
- **Connections:** microphone/brain-recorder wiring and any missing rack power cord are extra. Confirm the speaker's matching cable is included.
- **Before buying:** confirm the sound board fits the rack and has valid calibration. The driver bundle includes its enclosure and power supply; do not buy another. The starred output cable is our addition.

How precise is it?

- **Pitch range:** the speaker is specified for **4–110 kilohertz**. It misses the paper's lowest 1.4–4 kilohertz tones. The board's flat output extends to 92.1 kilohertz at 204,800 samples per second.
- **Start delay:** the sound board's output filter adds **0.15625 milliseconds** at its highest sample rate. Speaker and air delays are additional. The slides give no measured alignment with brain recording.
- **Drift:** the sound board's internal-clock specification corresponds to **72 milliseconds per hour versus ideal time** with current calibration; aging adds uncertainty. Clock settings and our recorder determine actual relative drift; that remains unknown.
- **Loudness at the ear:** accuracy is unmeasured. The speaker's uncorrected response variation is ± 11 decibels; that is not calibrated accuracy.

Sources: supplied diagram; supplied parts slide; sound board specifications, pages 19 and 24; speaker specifications; compatible board version. Supplier quote, shipping, tax, software and labor remain extra.

Accurate version 3: my recommendation

\$5,636.47 priced so far; speaker, speaker amplifier and powered enclosure need quotes.

What this buys

- Cover the paper's **1.4–32 kilohertz** range with a compact computer-connected sound device.
- Measure sound with a microphone near the ear. Record repeated timing signals with our brain data to align the recordings.
- This is a proposed build. Its sound level and timing have not been tested.

Parts to buy

United States dollars. One of each except batteries/resistors. Exact product model names are retained for ordering. Main subtotal uses a new microphone amplifier.

Part	Seller	Price
USB-6361 sound device, 781442-01	New / Newark	3,030.42
MF2-M speaker kit	Supplier quote	Quote
SA1 speaker amplifier	Supplier quote	Quote
ZB1PS powered enclosure	Supplier quote	Quote
Earthworks M50 measurement microphone [†]	New / eBay	1,299.00
Sound Devices MM-1 microphone amplifier [†]	New / B&H	945.00
REED R8090 sound-level calibrator [†]	New / DigiKey	219.00
Hosa HMIC-010 microphone cable [†]	New / Sweetwater	17.99
Gator RI-MICTP-FBM microphone stand [†]	New / Sweetwater	34.99
Two Panasonic microphone-amplifier batteries, \$0.62 each [†]	New / DigiKey	1.24
Pomona EM4970-36# output lead	New / DigiKey	23.68
Hosa PHX-106F-BULK microphone adapter	New / B&H	13.95
Two Yageo 100,000-ohm resistors, \$0.10 each	New / DigiKey	0.20
763830-01 United States power cord	New / Newark	51.00
Subtotal of priced parts		5,636.47

Ways to reduce cost / what is missing

- **Conditional used option:** earlier **\$325 MM-1**, no returns; subtotal **\$5,016.47** if available. Current availability is unconfirmed.
- **Shared measurement kit:** marked items total \$2,517.22 new, as used on pages 2 and 3; \$1,897.22 with the conditional used amplifier.
- **Still missing:** supplier quotes and the brain-recorder timing adapter. The microphone adapter needs rewiring before use.

How precise should we expect it to be?

- **Pitch and loudness:** the speaker covers 1–80 kilohertz; the microphone reaches 50 kilohertz. The sound-level calibrator is specified within **0.5 decibels at one kilohertz only**. Accuracy across all tones at the ear remains unmeasured.
- **Drift:** the independent clock specification corresponds to **180 milliseconds per hour versus ideal time**. Correct with repeated recorded references; remaining error is unmeasured.
- **Alignment goal:** sound-to-brain timestamps within **one millisecond after correction**. Untested; this does not describe the neural-processing delay measured by SONIC.

Included: device power supply/cable, microphone adapter and speaker stand/cable. Delivery/tax and labor are extra; no complete upper budget. Arduino or a field-programmable gate array still needs sound hardware.

Sources: [speaker](#); [microphone](#); [microphone amplifier](#); [sound-level calibrator](#); [sound-device timing](#). Clock figure is typical at 25 degrees Celsius; confirm computer support. Calculations: [supporting comparison](#).